

Template

Studentnames and studentnumbers here

2026-06-15

Set-up your environment

Title Page

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Include the tutorial group number Tutorial 5 Group 5

Include your tutorial lecturer's name Simon Loop

Use APA referencing throughout your document. Here's a link to some explanation.

Part 1 - Identify a Social Problem

The social problem being discussed is whether the growth of wages keeps up with the rising costs of living. The central question of this research is: does the growth of wages of the lowest-paid workers rise enough for them to be able to afford their current lifestyle? And does that apply across Europe?

1.1 Describe the Social Problem

- The social problem: The social problem is about the prices of living rising faster compared to wages, resulting in a worker's real purchasing power falling even if their wages rise on paper. This leads to workers being able to afford less each year.
- The relevance: The problem of the growing wages and the rising cost of living is relevant, because in recent years many countries in Europe have faced a sharp rise in prices, while the growth of wages falls behind, leaving many households with low-income struggling to financially keep up with the new living costs (ILO, 2022). The EU also implemented Directive (EU) 2022/2041 on adequate minimum wages in 2022, this aims to keep wage levels keep up with the growing inflation and living costs (European Parliament and Council of European Union, 2022).
- What the report will answer: The report will fill in the gap by comparing consumer-price inflation and minimum-wage levels across European countries throughout recent years. Existing reports look globally/nationally, but cross-country researches are under-examined, which is why this report will use cross-country methods to deepen such

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

This study's aim is to assess wages' rise in line with the country's rising living costs across Europe through the use of three datasets.

The Harmonised Index of Consumer Prices (HICP) was used for tracking rent/housing rates in Europe. The data were sourced from Eurostat and cover the annual rate of inflation for Europe up to the present (2022). The dataset also enables us to analyze the way that housing costs change over time (i.e., rent inflation rates from 2015 – 2025).

The second dataset used is based on the growth of the minimum wage rate in European countries as supplied by Eurostat. The statistical measures capture the annual increase in the statutory minimum wage for the countries in Europe that have a National Minimum Wage. This data allows for comparison of wage increases with others and helps determine if wage growth has been enough to keep up with price increases.

The third dataset used contains the geographical boundaries for European Countries that were supplied through the spatial package 'giscoR'. These geographical boundaries are used to illustrate rent inflation differences between the countries of Europe with maps.

When combined, these three individual datasets allow for comparison of how, both over time the cost to rent housing and minimum wage are related.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

Socioeconomic and economic indices, measured at the nation level, are the primary components of the datasets.

The rent data set comprises quantitative variables representing HICP (harmonised indices of consumer prices) values per year, which measure the change in consumer prices for housing-related items. This information allows analysts to assess how living costs have changed and how inflation is affecting households.

Wage datasets consist of quantitative variables that indicate how statutory minimum wage rates have changed from one year to another across the various countries. Labour force conditions and income levels among low-wage employees can therefore be analysed based on the available wage indices.

Country boundaries or country identifiers constitute the variables in the geographic dataset. While these geographic variables do not directly measure social or economic conditions, they allow for the geographical visualisation of socioeconomic indices.

The datasets originate from governmental administrations and official statistics rather than personal surveys or interviews. Economic data were gathered and developed by national statistical agencies and published by Eurostat. Thus, the datasets include standardised and equivalent figures across all European countries. we removed bosnia, turkey, and albania because some data was missing from them. *For the sake of this example, I will continue with the assignment...*

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

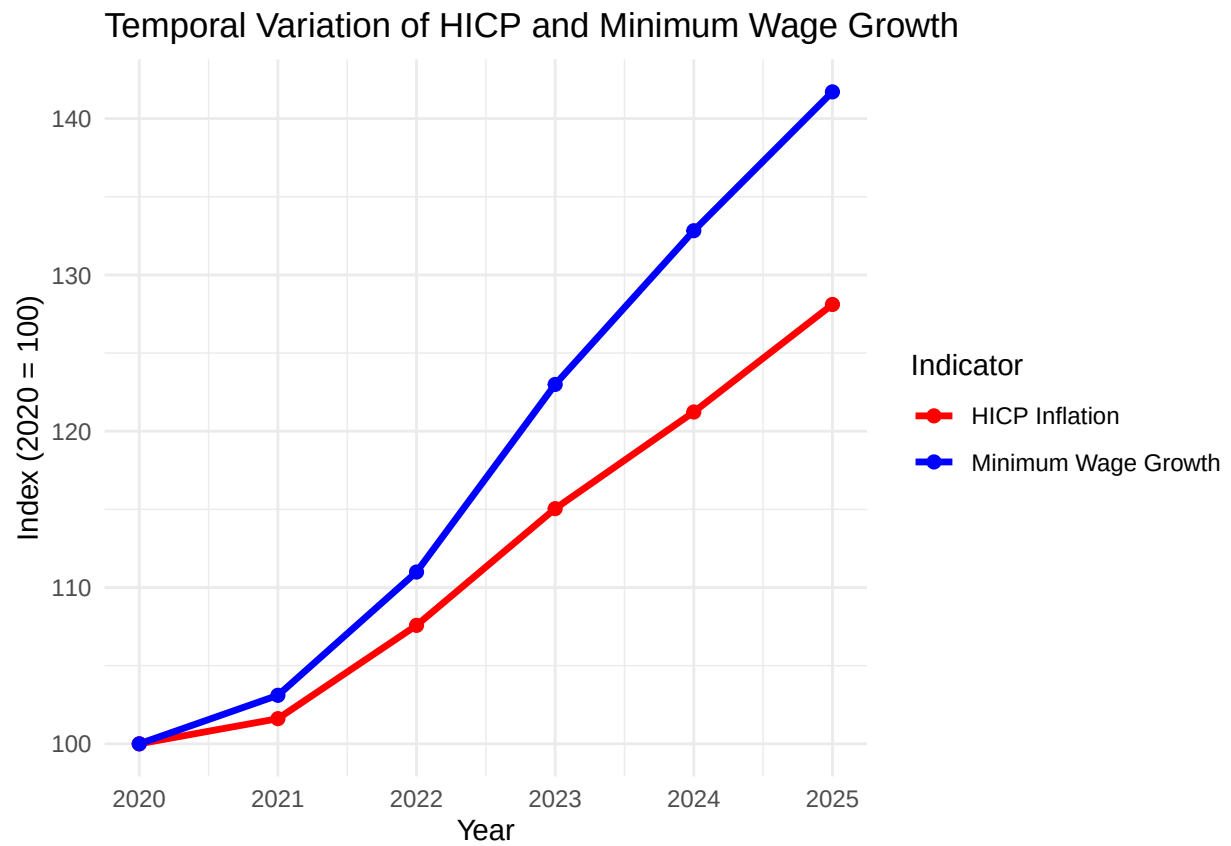
Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

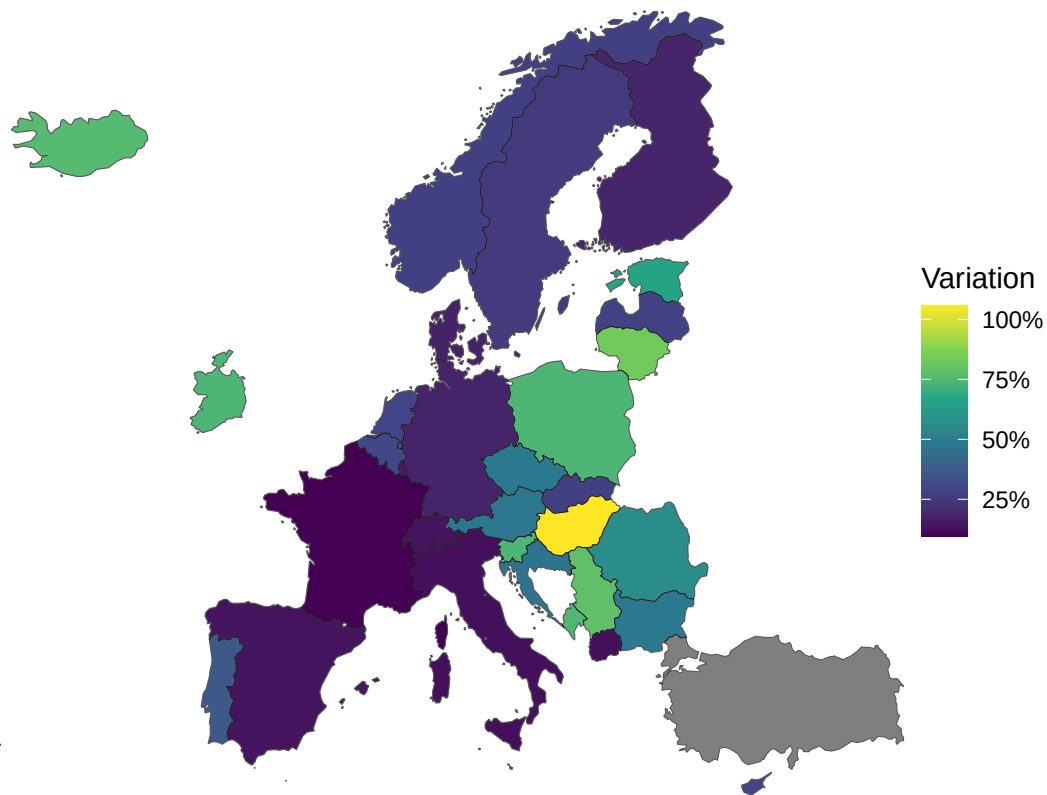
Variable 2

3.3 Visualize temporal variation



3.4 Visualize spatial variation

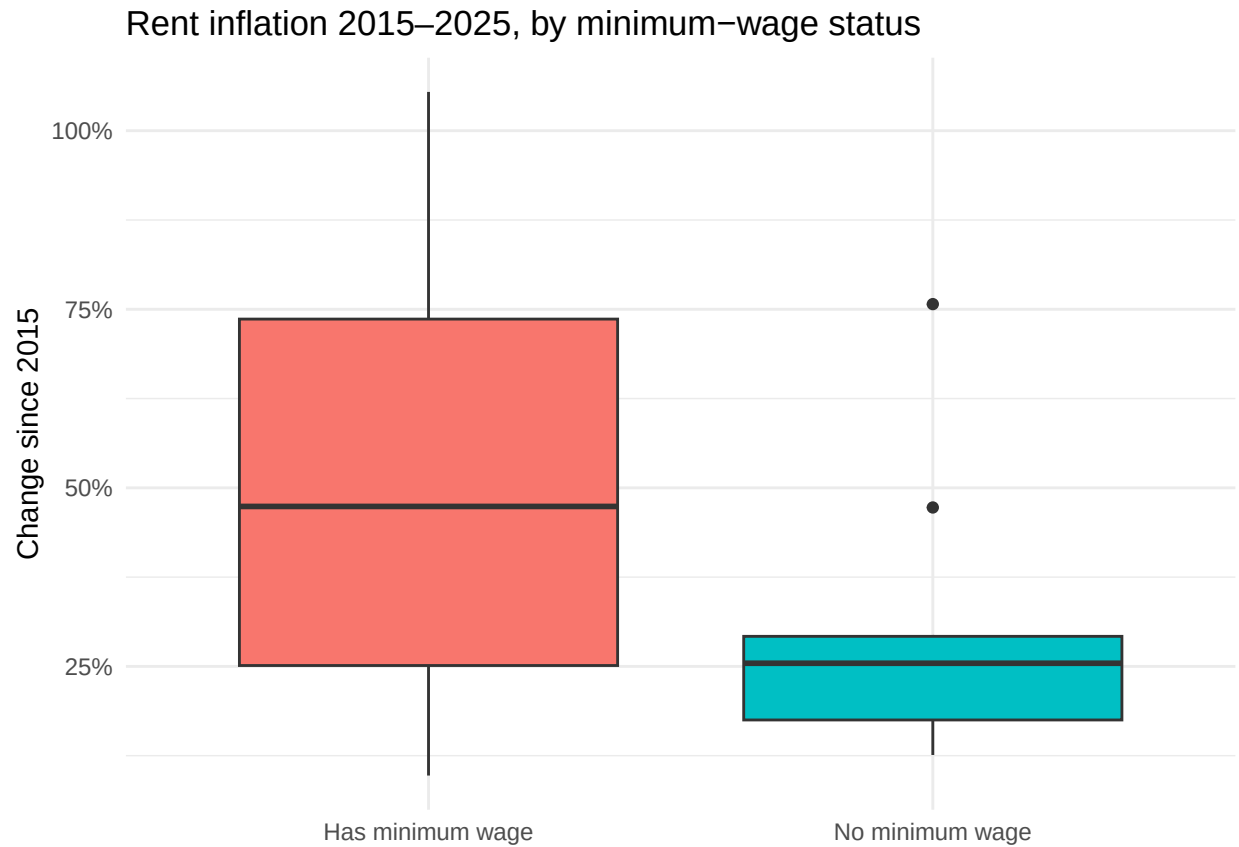
Map rent



Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

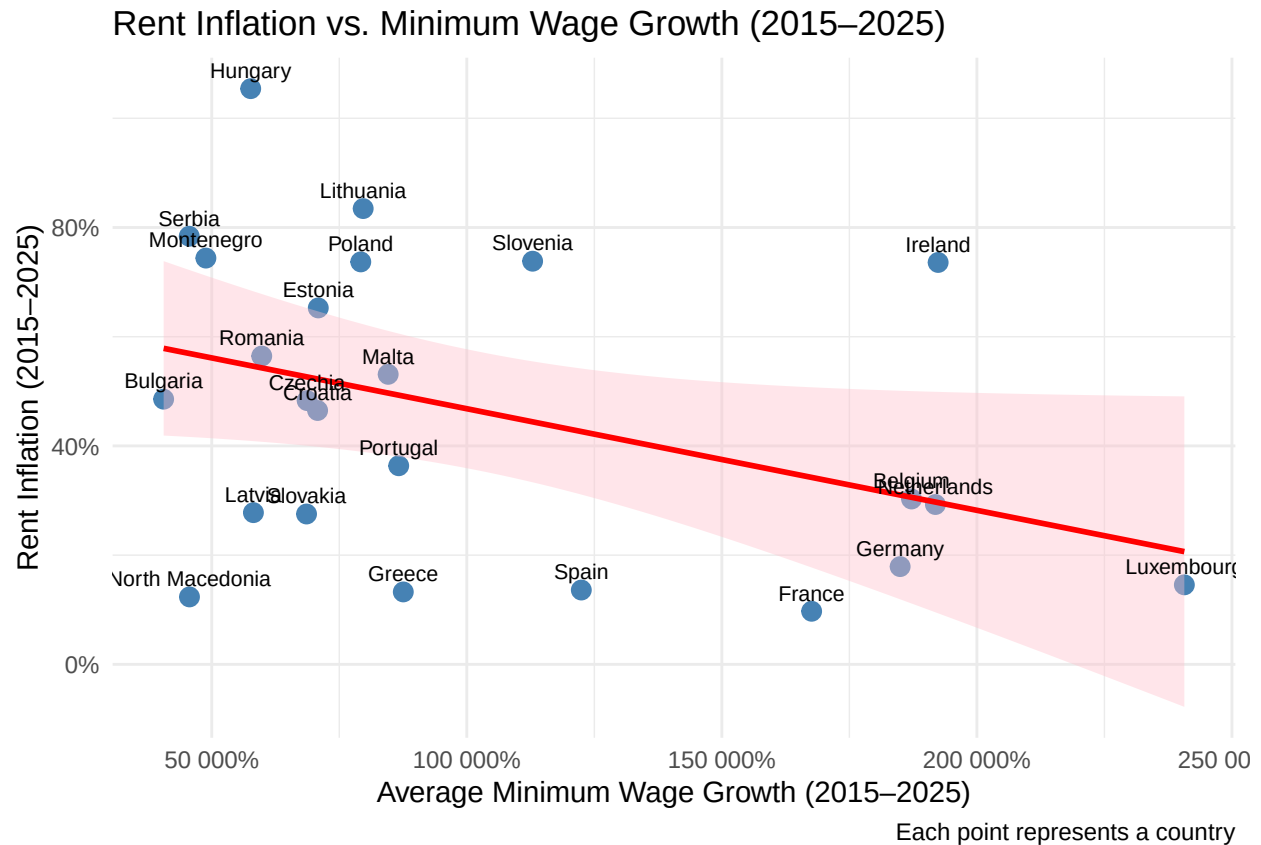


Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

```
## `geom_smooth()` using formula = 'y ~ x'
```



Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

International Labour Organization. (2022). Global Wage Report 2022–23: The impact of inflation and COVID-19 on wages and purchasing power. ILO. <https://www.ilo.org/publications/flagship-reports/global-wage-report-2022-23-impact-inflation-and-covid-19-wages-and>

Use APA referencing throughout your document.